

# Choptuik-QCD Bridge Verification Report

|                   |                          |
|-------------------|--------------------------|
| Author            | Ishak Khamzatovich Isaev |
| ORCID             | 0009-0003-7299-0701      |
| Generated         | 2026-08-10T14:02:13      |
| Elapsed (s)       | 0.001 s                  |
| Mode              | verify_all               |
| Sections executed | 5, 6, 7, 8, 9            |

## RESULTS

### *section\_5\_tau\_relax*

|                             |                                                                                                    |
|-----------------------------|----------------------------------------------------------------------------------------------------|
| theta_0                     | 1.000000e-19                                                                                       |
| tau_relax_s                 | 5.000000e-41                                                                                       |
| tau_relax_theory_s          | 3.291060e-24                                                                                       |
| Lambda_QCD_eV               | 200000000.000000                                                                                   |
| t_values_s                  | [60 items, first 5: [1e-45, 1.3141473626117527e-45, 1.7269832906594255e-45, 2.269510536694687e-45, |
| theta_t_values              | [60 items, first 5: [9.999800001999987e-20, 9.999737173981413e-20, 9.999654609306742e-20, 9.999546 |
| theta_at_1_tau              | 3.678794e-20                                                                                       |
| theta_at_5_tau              | 6.737947e-22                                                                                       |
| suppression_factor_at_1_tau | 0.367879                                                                                           |

### *section\_6\_kappa\_T\_physical*

|                              |                                                 |
|------------------------------|-------------------------------------------------|
| kappa_T_lower_95CL           | 2.620000                                        |
| kappa_T_best_fit             | 8.450000                                        |
| lattice_data_source          | Borsányi et al. arXiv:1512.04954 (extrapolated) |
| BF_at_lower                  | 99.000000                                       |
| BF_at_best_fit               | 510.000000                                      |
| BF_class_at_lower            | strong                                          |
| BF_class_at_best_fit         | decisive                                        |
| physical_kappa_in_GUE_regime | True                                            |
| GUE_threshold_kappa          | 1.500000                                        |

### *section\_7\_cabibbo*

|                        |          |
|------------------------|----------|
| b_Ch                   | 0.376510 |
| c_theta_framework      | 0.094128 |
| sin_2theta_C_predicted | 0.613604 |
| theta_C_predicted_rad  | 0.330309 |
| sin_theta_C_predicted  | 0.324335 |
| sin2_theta_C_measured  | 0.051000 |
| theta_C_measured_rad   | 0.227797 |

|                      |           |
|----------------------|-----------|
| sin_theta_C_measured | 0.225832  |
| deviation_rad        | 0.102512  |
| deviation_pct        | 45.001405 |
| coincidence_quality  | weak      |

## section\_8\_cp\_chain

### steps

[0]

|           |                                                                                 |
|-----------|---------------------------------------------------------------------------------|
| step      | 1                                                                               |
| statement | $O_{\chi} = Q_{\text{hat}}$ (structural role)                                   |
| evidence  | $O_{\chi}$ occupies the same epistemic niche as the topological charge operator |
| section   | §3                                                                              |

[1]

|           |                                                           |
|-----------|-----------------------------------------------------------|
| step      | 2                                                         |
| statement | $O_{\chi} = Q_{K3} \oplus M_F + \kappa_T * V_T$ at $N=28$ |
| evidence  | 22 $K3$ topological sectors $\oplus$ 6 quark flavors      |
| section   | §5.6                                                      |

[2]

|           |                                                                          |
|-----------|--------------------------------------------------------------------------|
| step      | 3                                                                        |
| statement | GUE class at $\kappa_T \geq 2.62$ (95% CL), $BF \geq 99$                 |
| evidence  | Bayes factor classification: strong at lower bound, decisive at best-fit |
| section   | §5.7, §6.4                                                               |

[3]

|           |                                                                 |
|-----------|-----------------------------------------------------------------|
| step      | 4                                                               |
| statement | GUE spectral symmetry $= \kappa_T \wedge \lambda \geq 0$        |
| evidence  | Wigner semicircle is symmetric; all odd spectral moments vanish |
| section   | §6.5                                                            |

[4]

|           |                                                                                |
|-----------|--------------------------------------------------------------------------------|
| step      | 5                                                                              |
| statement | Work formula: $\theta_{\text{bar}} = \delta_C * N * \kappa_T * S_{\text{GUE}}$ |
| evidence  | Derived from path integral over topological sectors                            |
| section   | §6                                                                             |

[5]

|           |                                                           |
|-----------|-----------------------------------------------------------|
| step      | 6                                                         |
| statement | $\theta_{\text{bar}} = 0$ exactly in continuum GUE regime |
| evidence  | Follows directly from steps 4 and 5                       |
| section   | §6                                                        |

[6]

|           |                                                                           |
|-----------|---------------------------------------------------------------------------|
| step      | 7                                                                         |
| statement | Finite- $N$ artifact $\sim 1/\sqrt{N}$ vanishes as $N \rightarrow \infty$ |
| evidence  | Monte Carlo verification across $N = 10..10000$                           |

|         |      |
|---------|------|
| section | §6.6 |
|---------|------|

[7]

|           |                                            |
|-----------|--------------------------------------------|
| step      | 8                                          |
| statement | Dynamic relaxation tau_relax ~ 5e-41 s     |
| evidence  | Damps CKM-induced residual theta_0 ~ 1e-19 |
| section   | §6.7                                       |

|                           |                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------|
| total_steps               | 8                                                                                                                |
| final_result              | theta_bar = 0 exactly                                                                                            |
| new_fields_introduced     | 0                                                                                                                |
| new_scales_introduced     | 0                                                                                                                |
| new_symmetries_introduced | 0                                                                                                                |
| falsification_tests       | ['Direct lattice measurement of F(theta) - F(0) via Giusti-Rossi-Testa method', 'Derivation of work formula from |

## section\_9\_jet\_wake

|                          |                                          |
|--------------------------|------------------------------------------|
| delta_C                  | 0.448799                                 |
| Lambda_QCD_GeV           | 0.200000                                 |
| chi_eff_GeV4             | 7.180783e-04                             |
| chi_eff_eV4              | 718078320820524292425978688307200.000000 |
| bridge_formula           | chi_eff = delta_C * Lambda_QCD^4         |
| jet_wake_amplitude_ratio | 0.142857                                 |
| topological_sector_count | 22                                       |
| kappa_T_coupling         | 8.450000                                 |

## EXECUTION LOG

```
[2026-08-10 14:02:13] Starting QCD bridge run, mode=verify_all, sections=[5, 6, 7, 8, 9]
[2026-08-10 14:02:13] Language: en
[2026-08-10 14:02:13] Output dir: reports
[2026-08-10 14:02:13] Report formats: ['txt', 'csv', 'md', 'pdf', 'html', 'docx', 'json']
[2026-08-10 14:02:13] Section 5: tau_relax dynamics
[2026-08-10 14:02:13]   tau_relax = 5.00e-41 s
[2026-08-10 14:02:13] Section 6: kappa_T physical estimate
[2026-08-10 14:02:13]   kappa_T lower 95%CL = 2.62, BF = 99.0
[2026-08-10 14:02:13] Section 7: Cabibbo angle coincidence
[2026-08-10 14:02:13]   theta_C predicted = 0.330309, measured = 0.227797, deviation = 45.00%
[2026-08-10 14:02:13] Section 8: CP 8-step solution chain
[2026-08-10 14:02:13]   Chain complete: 8 steps -> theta_bar = 0 exactly
[2026-08-10 14:02:13] Section 9: Jet wake bridge
[2026-08-10 14:02:13]   chi_eff = 7.180783e-04 GeV^4
[2026-08-10 14:02:13] QCD bridge run complete in 0.001s
```